2024 IEEE International Conferences on Internet of Things (iThings) and IEEE Green Computing & Communications (GreenCom) and IEEE Cyber, Physical & Social Computing (CPSCom) and IEEE Smart Data (SmartData) and IEEE Congress

2024 IEEE International Conferences on Internet of Things (iThings) and IEEE Green Computing & Communications (GreenCom) and IEEE Cyber, Physical & Social Computing (CPSCom) and IEEE Smart Data (SmartData) and IEEE Congress on Cybermatics | 979-8-3503-5163-7/24/$31.00 ©2024 IEEE | DOI: 10.1109/iThings-GreenCom-CPSCom-SmartData-Cybermatics62450.2024.00104

STIOS: A Novel Self-supervised Diffusion Model for Trajectory Imputation in Open Environment Scenarios

1st Zhijing Hu
Laboratory for Big Data Data and Decision National University of Defense Technology
Changsha, China
huzhijing@nudt.edu.cn

2nd Hao Yan
School of Computer Science and Engineering Central South University
Changsha, China
CSUyh1999@csu.edu.cn

3rd Yuhao Zheng
School of Computer Science and Engineering Central South University
Changsha, China
8212200810@csu.edu.cn

4th Hua He
Laboratory for Big Data Data and Decision National University of Defense Technology, China
hehua3566@nudt.edu.cn

5th Chao Chen
Laboratory for Big Data Data and Decision National University of Defense Technology, China
chenc1997@nudt.edu.cn

6th Changjun Fan*
Laboratory for Big Data Data and Decision National University of Defense Technology, China
Changjun Fan@nudt.edu.cn

7th Kuihua Huang*
Laboratory for Big Data Data and Decision National University of Defense Technology, China
khhuang@nudt.edu.cn

***Abstract*—With the growth of location-based services, the accumulation of large amounts of trajectory data comes with the challenge of missing data. Existing trajectory imputation methods rely on deterministic models that cannot take into account open environments scenarios without auxiliary information. This may lead to poor performance in learning the long-term dependencies of trajectories. In this paper, we propose a novel Self-supervised diffusion model for Trajectory Imputation in Open environment Scenarios (STIOS), which combines a deterministic model and a conditional diffusion model. In addition, we propose a self-supervised training method inspired by masked language modelling that divides the data into conditional information and estimation targets. Extensive experiments on two real datasets demonstrate the superiority of STIOS for trajectory imputation.**

***Index Terms*—Diffusion model, Trajectory imputation, Open environment scenarios**

I. Introduction

The collection and utilization of location data recently have become ubiquitous, providing unprecedented opportunities to optimize our living spaces and experiences. However, this technological advancement is coupled with challenges to data integrity and quality, with data loss due to device failure or human error. And various other reasons being inevitable and significantly detrimental to the performance of downstream applications. In this study, we focus on the problem of imputation on spatio-temporal trajectory data, where the trajectory imputation task aims to predict and fill in missing values in the trajectory data. Specifically, this task typically involves giving a segment of incomplete trajectory data in the form of two-dimensional data. And then the associated imputation model is required to fill in all the trajectory data at the missing locations based on the input. Unfortunately, so far, there is still a lack of promising solutions to trajectory imputation problems [1]. Efforts have been made in trajectory data imputation methods which are typically based on time series data imputation. In recent years, methods have evolved from early models to deep learning deterministic models. For trajectory imputation, [2], [3] designed multiple attention mechanisms to capture the periodic patterns of human movement. [4], [5] utilized graph neural networks for trajectory recovery and there are lots of works that focus on trajectory representation learning(TRS) [6], [7].

However, the problem addressed is more challenging compared to previous work for the following reasons. Existing trajectory imputation models are mainly targeted at traffic trajectories [8]–[13]. Since these traffic trajectories usually have road network, or road information, with many constraints, the difficulty of trajectory imputation performed in this way is much easier. In contrast, it is difficult for these existing models to deal with trajectories without a priory information in open environment scenarios. Examples include bird flight trajectories, human walking trajectories, and low-altitude Unmanned Aerial Vehicle (UAV) trajectories. These trajectories lack auxiliary information, as shown in “Fig. 1”. We consider that any additional data information other than the trajectory data itself can be considered as auxiliary or a priory information, but it is lack of predetermined paths and constraints in open environment scenarios. And we need

2836-3701/24/$31.00 ©2024 IEEE

DOI 10.1109/iThings-GreenCom-CPSCom-SmartData-Cybermatics62450.2024.00104

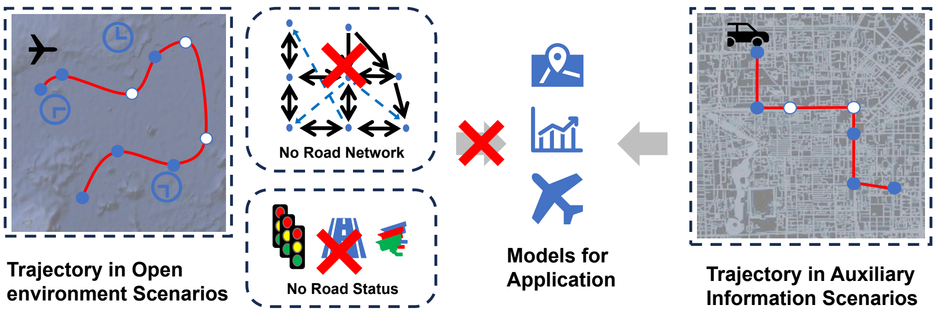

Fig. 1: Trajectory in different information scenarios. At this point, many models do not perform well in open environment scenarios, because there is no access to valid pavement information inputs, such as road network or road state.

promising solutions to these imputation challenges. In addition, trajectory data is a special type of spatio-temporal data with specific positional relationships and long-term dependencies in time series modelling. Previous methods cannot learn this well, e.g., Graph Neural Network (GNN), is more suitable for capturing static relationships. Therefore, a new method is needed to capture complex temporal and dynamic spatial dependencies more efficiently. In addition, trajectory data formats are inconsistent and the amount of valid data is small. The variety of data in the trajectory data domain has led to a lack of uniform data standards in the field due to privacy or commerciality. This often lacks effectively labelled trajectory data for imputation tasks, limiting the conditions for supervised learning. Therefore, new training methods need to be developed to capture the intrinsic patterns of the data.

To address the above challenges and the shortcomings of existing models, this paper proposes STIOS for trajectory imputation in open environment scenarios. Specifically, we combine the diffusion model with a deterministic model to incorporate long-term dependencies. Furthermore, due to the small amount of data in the model and the lack of labels, we develop a self-supervised training method inspired by masked language modelling for training the model. In summary, our contributions are as follows:

- We propose STIOS model for trajectory imputation with specific positional relationships and long-range dependencies in open environment scenarios.
- We propose a self-supervised training method inspired by masked language modeling, dividing the data into condition and imputation targets.
- We provide extensive experimental evidence, where numerous experiments demonstrate the strong competitiveness of our STIOS model and its superiority in achieving accurate trajectory imputation.

II. Related Work

Missing values are common in trajectory data and multivariate time series data. To fully utilize and analyze them, the related research has become a very rich topic [14], [15].

How to impute trajectory has always been an important issue in spatiotemporal data analysis and management, with most work modeling using sequence learning models. [2] modeled the periodicity of mobility learned from history with Recurrent Neural Networks (RNNs), and [3] designed various attention mechanisms to capture the periodic patterns of human movement. Additionally, [3], [5], [16] also utilized graph neural networks to capture complex locational transition patterns for trajectory recovery. [17] proposed a non-autoregressive multi-resolution sequence imputation model to establish non-autoregressive correlations between missing trajectories and temporally adjacent observations. [18] used Long Short-Term Memory model (LSTM) encoding and decoded using hidden features for trajectory imputation for each object. However, they are only suitable for scenarios with strong information,based on road networks,and do not perform well in open environment scenarios.

Deterministic models can capture the temporal dependencies of time series and have shown their excellent performance, which performs representation learning and modeling of sequences to generate determined imputations given observed values. [19]introduced bidirectional recurrent neural networks to effectively estimate missing values. [20] and [21] replace the missing values in the input sequence with specific markers. To enhance the capability of representation learning, [22] applied a self-training mechanism to multivariate imputation, and [23] proposed attention-based multivariate imputation models, considering the inter-variable correlations. [24] employed self-attention to capture time features and feature correlations.

Given the importance of accounting for uncertainty in various real-world scenarios [15], [25], there has been an increasing effort focused on probabilistic methods in recent years. Early attempts at probabilistic models for Missing Data Imputation expanded Generative Adversarial Networks (GANs) [26], [27] to multivariate imputation, which resulted in a proliferation of GAN variants each addressing different aspects of the problem, such as E2-GAN [28] and SSGAN [29]. Recently, diffusion models [30], [31] have shown impressive performance in various generative tasks, such as audio and image synthesis. [32] attempted to learn the conditional distribution of diffusion models based on conditional scores, [14] takes into account the correlation and ensuring consistency between observed values and missing values.

In summary, these above methods either neglect the spatiotemporal data characteristics of trajectory data , or are based on road networks proposing various spatial correlation construction methods. Therefore, imputation for deficient information scenarios remains an existing issue with current models still needing improvement.

III. Preliminary

In this paper, we focus on the problem of trajectory sequence imputation in open environments. Our goal is to estimate the missing trajectory values based on the observed values of these sequences. The trajectory sequences we are interested in contain several sequences of the same length, each of which contains missing values. We here formally define the problem of trajectory imputation [14].

The processed trajectory sequences containing N trajectories of equal length L can be represented as X, where X^{cond} is the observable trajectory data, X^{miss} is the missing trajectory data, and $\mathcal{M}$ is the missing record matrix, $\mathcal{M} \in [0,1]^{N\times L}$, where X^{cond} and X^{miss} are defined as the subsets of X corresponding to observed and missing data, respectively.

We represent the i-th trajectory as X^i, and considering that other additional information may be missing in general scenarios, we only consider the longitude and latitude values $x^{i,t}_{lon}, x^{i,t}_{lat}$ for each trajectory point $x^{i,t}$, assuming that the longitudinal and latitudinal data are correlated.

Taking into account the practical issue of random missing trajectory information, we uniformly randomly sample a proportion α of points from the all-ones matrix $\mathbf{1} \in \mathbb{R}^{N\times L}$ and assign a value of 0 to the corresponding positions to obtain the missing record matrix $\mathcal{M} \in [0,1]^{N\times L}$, thus taking $X, \mathcal{M}, \mathcal{N}$ as input. We define the problem of trajectory sequence imputation in open environments as training a network model p_θ to approximate the true distribution $q(X)$:

$$\max_\theta p_\theta(X^{miss}|X^{cond}) \tag{1}$$

where $(X^{miss}, X^{cond}) \sim q(X)$ and p_θ is a probabilistic imputation model aimed at approaching the true conditional distribution $q(X^{miss}|X^{cond})$ [14].

IV. Method

A. Transformer-based trajectory imputation diffusion model

1) Imputation diffusion model: Denoising diffusion probabilistic models have demonstrated state-of-the-art performance for multivariate time series imputation [32]. Based on the premise that the diffusion process is a Markov process, the diffusion model aims to use the learned distribution $p_\theta(X)$ to approximate the true distribution $p(X)$, learning to map from latent space to original signal space by removing noise in the reverse process that is added sequentially in a Markovian manner during the forward process [15].

The forward noising process can be parameterized as:

$$q(X_t|X_{t-1}) = \mathcal{N}(X_t; \sqrt{1-\beta_t}X_{t-1}, \beta_t I) \tag{2}$$

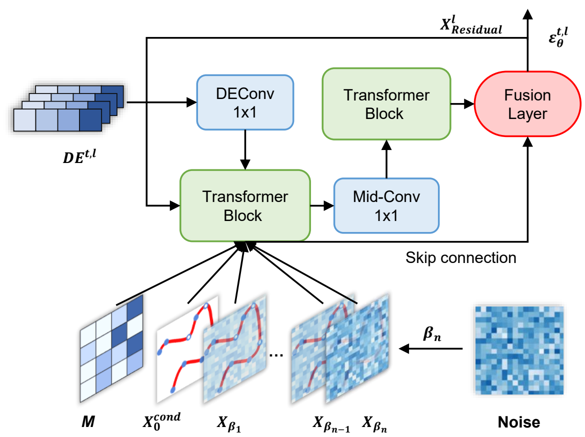

Fig. 2: The architecture of STIOS, where the masked trajectory data, after being noised at step t, and diffusion embedding $DE^{t,l}$ to output the step t predicted noise $\epsilon^{t,l}_\theta$ at layer l. The outputs from all layers l are compiled to form the reconstructed noise ϵ_θ

where β_t represents the noise level, that is, given x_{t-1}, then $x_t = \sqrt{1-\beta_t}x_{t-1}$. Given the original sequence x_0, we can ultimately obtain the noised data x_t. We replace α_t with $1-\beta_t$, yielding:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon \tag{3}$$

where $\epsilon \sim \mathcal{N}(0, I)$ and $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$.

The reverse denoising process can be parameterized as:

$$\begin{aligned} &p(X_{t-1}|X_t, X_0) = \mathcal{N}(X_{t-1}; \mu(X_{t-1}|X_t, X_0), \sigma(t)), \\ &s.t.\ \mu(X_{t-1}|X_t, X_0) = \frac{1}{\sqrt{\alpha_t}}X_t - \frac{1-\alpha_t}{\sqrt{\alpha_t(1-\bar{\alpha}t)}}\epsilon \end{aligned} \tag{4}$$

where $\epsilon \sim \mathcal{N}(0, I)$, and the variance $\sigma(t)$ is solely dependent on t, specified as:

$$\sigma(t) = \frac{1-\bar{\alpha}t-1}{1-\bar{\alpha}t}(1-\alpha_t). \tag{5}$$

At this point, given x_0 and x_t, the network model outputs ϵ_θ, denoising x_t into x_{t-1}, that is:

$$X_{t-1} = \frac{1}{\sqrt{\alpha_t}}X_t - \frac{1-\alpha_t}{\sqrt{\alpha_t(1-\bar{\alpha}t)}}\epsilon\theta(X_t, t) \tag{6}$$

To minimize the Kullback-Leibler (KL) divergence between $q_\theta(X^{miss}_{t-1}|X_t)$ and the true distribution $q(X^{miss}_{t-1}|X_t)$, [32] demonstrated that the reverse process can be trained by solving the following optimization problem:

$$\min_\theta E_{X_0\sim q(x_0),\epsilon\sim\mathcal{N}(0,I),t}||(\epsilon - \epsilon_\theta(x_t, t))||_2^2 \tag{7}$$

2) STIOS model architecture: The proposed model framework, as shown in "Fig. 2", we have studied numerous diffusion imputation architectures, and current generative spatio-temporal data imputation models are largely based on Variational Auto-Encoder (VAEs) or GANs, with no prior work directly applying conditional diffusion models to spatio-temporal data imputation. Considering the similarities between spatio-temporal data and time series data, we introduce time series imputation models to spatio-temporal data imputation. Some progress has been made on this issue by predecessors, for instance, the previously mentioned CSDI [32], SSSD^{S4} [15], or MIDM [14], all of which are time series imputations based on the DiffWave framework. Therefore, we attempt to adapt time series imputation models from the DiffWave architecture as a baseline for trajectory sequence completion and propose a more suitable completion model for general trajectory data.

STIOS includes two Transformer Layers, multiple one-dimensional convolutional layers, and processed trajectory data. First, the trajectory data, post-masking, undergoes varying degrees of noise addition, and the Diffusion Embedding at step t is computed, and the trajectory data for this portion pass through convolution layer to obtain embedding which is inputted into the network architecture.

Finally, the data is processed centrally in the Fusion layer, while simultaneous residual connections are executed, redirecting the output back as input for the next step. In " IV-A1", we introduce the two principal processes involved in the STIOS framework. The noise-adding process is designed to iteratively introduce noise into the missing trajectory sequences, which eventually converge to a Gaussian distribution. The denoising process reconstructs the Gaussian noise at each step, ultimately yielding the imputed data for the trajectory sequence. At each step of the reverse trajectory denoising process, the noise is amalgamated by the fusion layer, resulting in the final predictive noise.

B. Implementation of STIOS for Imputation

1) Self-supervised Training: We employ a training concept similar to the masked language modeling used in natural language processing, using M for artificially missing record matrices, where $M \in [0,1]^{N \times L}$. Through the missing record matrix, we divide into conditional data X^{cond} and imputation data X^{miss} according to the assumptions in section 3.1, where $X^{cond} = X^{i,t}|M^{i,t} = 1$ and $X^{miss} = X^{i,t}|M^{i,t} = 0$. During training, we can follow the training process of the unconditional model in section 4.1.1, adding X_0^{cond} from the map, which makes the optimization target become:

$$\min_{\theta} E_{X_0 \sim q(x_0), \epsilon \sim N(0,1), t} ||(\epsilon - \epsilon_\theta(x_t, t, x_0^{cond}))||_2^2 \quad (8)$$

In this context, ϵ should be consistent with the dimension of the input X_0^{cond}. During training, we treat the fabricated X^{miss} as missing values of the trajectory for training, which are set to 0 or NaN. The specific algorithm can be seen in Algorithm 1, and the training process based on mask language modeling concept is shown in "Fig. 3".

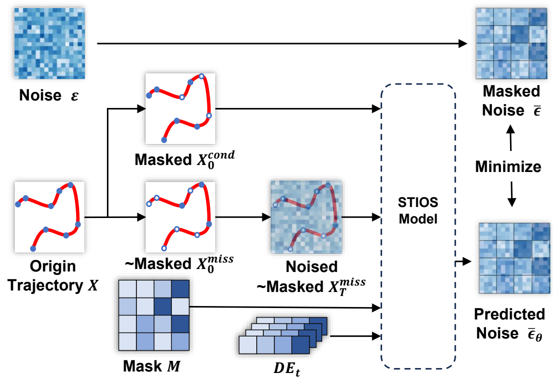

Fig. 3: The self-supervised masked training process is inspired by masked language model,which dividing the data into conditional information and imputation target. We stack the the step t diffusion embedding $DE^{t,l}$ and predicted noise $\epsilon_\theta^{t,l}$ of all l layers in Figure 2 to represent DE_t and $\bar{\epsilon_\theta}$.

Algorithm 1 Training of STIOS

Require: Distribution of training data $q(x_0)$, number of iterations $\mathcal{N}_{iter}$, sequence of noise levels $\{\alpha_t\}$, and integrated trajectory data $\{X, \mathcal{M}, \mathcal{N}\}$

Ensure: Trained denoising function ϵ_θ

for $i = 1$ to $\mathcal{N}_{iter}$ **do**
  $t \sim Uniform(\mathbf{1}, ..., T), x_0 \sim q(x_0)$
  $X^{cond} = X \odot M, X^{miss} = X \odot (1-M)$, $\epsilon \sim N(0, I)$, where ϵ and X have the same dimensions
  Treat X^{cond} and X^{miss} as $X_{\mathbf{0}}^{cond}$ and $X_{\mathbf{0}}^{miss}$, respectively, and add noise to $X_{\mathbf{0}}^{miss}$ to get $x_t^{miss} = \sqrt{\bar{\alpha}_t} x_{\mathbf{0}}^{miss} + \sqrt{\mathbf{1} - \bar{\alpha}_t}\epsilon$
  Perform a gradient step according to Equation 9:$\nabla_\theta ||(\epsilon - \epsilon_\theta(x_t, t, x_0^{cond}))||_2^2$
end for

2) STIOS Imputation: During the imputation inference, we initially sample random noise ϵ from a standard Gaussian, denoted as x_T, and through the reverse process of the conditional diffusion model, at each time step t, we utilize the $\epsilon_\theta(X_t, t, X_{t-1}^{cond})$ generated by the trained network using the given missing data x_0^{cond}. This allows us to obtain $X_{t-1}^{miss(imp)}$, progressively denoising to ultimately produce the generated imputation data $X_0^{miss(imp)}$, gradually transforming noise into a plausible completed trajectory sequence. The specific Algorithm 2 is shown below.

V. Experiment

A. Experiment Setup

In this section, we demonstrate the efficacy of STIOS for the imputation of trajectory sequences. Four datasets were employed, and our methodology is explicated in three facets concerning the construction of the datasets and evaluation metrics.

Algorithm 2 Imputation of STIOS

Require: Distribution of training data $q(x_0)$, number of iterations $\mathcal{N}_{iter}$, sequence of noise levels $\{\alpha_t\}$, integrated trajectory data $\{X, \mathcal{M}, \mathcal{N}\}$,Trained denoising function ϵ_θ, and number of sampling n_{sample}

Ensure: Imputation data $X_{\mathbf{0}}^{miss(imp)}$

$X^{cond} = X \odot M, \epsilon \sim N(\mathbf{0},\ I)$, where ϵ as $X_T^{miss(imp)}$, and have the same dimensions with X

for $i = T$ to 1 **do**

  $t \sim Uniform\,(\mathbf{1}, ..., T)\,, x_0 \ \sim\ q(x_{\mathbf{0}})$

  $X^{cond} = X \odot M, X^{miss} = X \odot (1 - M),\ \epsilon \sim N(0, I)$, where ϵ and X have the same dimensions

  Treat X^{cond} as $X_{\mathbf{0}}^{cond}$, and denoise to $X_T^{miss(imp)}$ to get $X_{t-\mathbf{1}}^{miss(imp)} = \frac{\mathbf{1}}{\sqrt{\alpha}_t} X_t - \frac{\mathbf{1} - \alpha_t}{\sqrt{\alpha_t(\mathbf{1} - \bar{\alpha}_t)}} \epsilon_\theta(X_t, t)$

  generating multiple Imputation(Sampling) with hyper-parameter n_{sample}

end for

1) Dataset: We evaluated the performance of the proposed model on two taxi trajectory datasets.

- **Geolife**[1]**.** Geolife is collected from April 2007 to August 2012 by 182 users for a geospatial project at Microsoft Research Asia, and represents the urban scenario. The GPS trajectories in the dataset consist of sequences of timestamped points, including information on latitude, longitude, and altitude. Such datasets are extensively used for tasks such as trajectory prediction and travel time estimation.
- **LocaRDS**[2]**.** LocaRDS is an open reference dataset of real-world crowdsourced flight data from the OpenSky Network, containing more than 500 million measurements from over 50 million transmissions recorded by 323 sensors. This dataset is used for aerospace scenarios and can be employed for analysis in spatio-temporal data tasks.

The entire datasets were partitioned into non-overlapping training, testing, and validation sets with a ratio of 8:1:1. Effective datasets were compiled by segmenting and filtering valid trajectories, with detailed data presented in Table I.

TABLE I: Statistics of the 2 Datasets After Preprocessing

Dataset	Location	Scenarios	Duration	Traj Pair
Geolife	Beijing	Urban Traffic	5 years	993
LocaRDS	Open Sky	Airspace	3600 seconds	921

2) Data Preprocessing: We used three diverse methods of data processing, as follows:

- **Data Deformation.** We employed the Min-Max scaling technique to constrain the data range, as we discerned that trajectory data are particularly sensitive to precision. The use of common nonlinear transformations significantly affects the outcomes, as minor decimal variances in latitude and longitude during transformation can lead to substantial positional discrepancies upon data restoration, thereby considerably impacting the numerical results. Fine-tuning based on the Min-Max transformation allowed us to concentrate on the precision of the decimal points, effectively resolving this issue.
- **Data Missing and Cutting.** Reflecting the real-world challenge of trajectory data being incomplete, we embraced the concept of random point omission, designating 10% of the trajectory points within a region as missing, effectively setting all corresponding information to zero. Concurrently, to standardize our data, we confined each trajectory to a maximum length of l_{max}. Trajectories exceeding l_{max} were trimmed to l_{max} points, while trajectories shorter than l_{max} were padded with zeroes and their positions marked using a padding matrix $Mask_{Pad}$. Trajectories with fewer than l_{min} points were discarded. For longer trajectories, we sampled points at consistent time intervals to limit their length to l_{max}. In this study, we set $l_{max} = 200$ and $l_{min} = 50$, referring to [32].
- **Adding Noise.** Determining the appropriate level of Gaussian noise, β, to introduce into the data has always been an intriguing area of exploration. We established four different schedules during our experiments: cosine, sigmoid, quadratic, and linear. In our experiments, we opted for a quadratic schedule to set β unevenly, achieving favorable results. We established an initial noise level $\beta_{start} = 0.0001$ and increased it to a final level of $\beta_{end} = 0.5$.

3) Baseline: In scenarios where information loss is prevalent and conventional models reliant on road network data are inadequate for imputation, we benchmark our proposed STIOS model against several widely-used models for multidimensional time series imputation, as well as state-of-the-art models. We consider both deterministic and generative models for a comprehensive comparison.

Deterministic Models

- KNN [33]: This model uses the average of the k-most similar variables for imputation; we set k=10 in our experiments.
- Transformer Model [34]: Leverages attention mechanisms to impute serialized data, catering to the sequential nature of time series.
- BRITS [21]: An RNN-based approach that utilizes recurrent dynamics to estimate missing values in multivariate time series.
- SAITS [24]: Learns the missing values from a weighted combination of two Diagonal Masked Self-Attention (DMSA) blocks.

Probabilistic Models

- GP-VAE [35]: A deep probabilistic model for imputing multivariate time series that integrates concepts from

[1]https://www.microsoft.com/en-us/download/details.aspx?id=52367
[2]https://zenodo.org/records/4302265

TABLE II: Performance Comparison for Deterministic & Probabilistic Models[a]

Dataset		Geolife			LocalRDS		
Models Type	**Method**	**GridACC**	**RMSE**	**MAE**	**GridACC**	**RMSE**	**MAE**
Deterministic	KNN-10	0.1533(0.0006)	8.0485(0.0244)	5.1012(0.0549)	0.0177(0.0002)	0.2203(0.0055)	0.1608(0.0021)
	Transformer	0.0281(0.0003)	10.212(0.2653)	6.2554(0.1457)	0.0014(0.0001)	0.1725(0.0043)	0.1440(0.0030)
	BRTIS	0.6498(0.0072)	1.7892(0.0231)	0.8859(0.0110)	0.2170(0.0028)	0.0631(0.0017)	0.0226(0.0005)
	SAITS	0.7186(0.0056)	3.7011(0.0664)	0.9635(0.0189)	0.4343(0.0049)	0.0908(0.0022)	0.0167(0.0003)
	STIOS	**0.9060**(0.0033)	**0.4561**(0.0122)	**0.1801**(0.0044)	**0.9475**(0.0018)	**0.0020**(0.0001)	**0.0079**(0.0002)
Probabilistic	GP-VAE	0.0705(0.0009)	13.346(0.0321)	10.233(0.0468)	0.0010(0.0001)	0.1764(0.0039)	0.1489(0.0035)
	SSSD	0.0677(0.0008)	5.1085(0.0204)	4.1287(0.0302)	0.0020(0.0002)	0.2133(0.0046)	0.1698(0.0028)
	USGAN	0.6425(0.0081)	2.1580(0.0257)	0.9195(0.0103)	0.2000(0.0025)	0.0603(0.0016)	0.0232(0.0004)
	CSDI	0.8766(0.0044)	0.5947(0.0135)	0.2212(0.0053)	**0.9533**(0.0020)	0.0024(0.0001)	**0.0064**(0.0002)
	STIOS	**0.9060**(0.0033)	**0.4561**(0.0122)	**0.1801**(0.0044)	0.9475(0.0018)	**0.0020**(0.0000)	0.0079(0.0002)

[a]Bold text is the best data, and underlined data is the second best.

Variational Autoencoders (VAE) with Gaussian Processes (GP).

- SSGAN [29]: Viewed as a GAIN [36] variant with bidirectional recurrent encoders and decoders.
- CSDI [32]: A recent standout in diffusion models, it employs a conditional diffusion model and uses the Diffwave architecture for estimating missing values in multivariate time series.
- SSSD[S4] [15]: Adapts recent advances in state-space models as denoising modules, combining them with Diffwave to facilitate imputation.

These models are evaluated under similar conditions to ensure a fair comparison. For each model, specific parameters are adjusted to optimize performance on the task of imputing missing trajectory sequences. The deterministic models tend to offer predictions based on learned patterns without randomness, whereas the generative models provide a probability distribution from which multiple plausible imputed values can be drawn. The choice between deterministic and generative models may depend on the specific application and whether uncertainty quantification of the imputed values is necessary.

4) Evaluation Metrics: To rigorously evaluate the performance of models in imputing missing data, we propose three metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Grid Prediction Accuracy (GridACC). These metrics serve as quantitative measures of prediction accuracy, each capturing different aspects of the imputation quality. The formulas and their purposes are outlined below:

- **Root Mean Square Error (RMSE).** RMSE measures the square root of the average squared differences between the predicted values ($\mathcal{X}^i$) and the true values ($\mathcal{Y}^i$). A lower RMSE indicates better predictive accuracy.

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(|\mathcal{X}^i - \mathcal{Y}^i|_F^2)} \tag{9}$$

- **Mean Absolute Error (MAE).** MAE calculates the average of the absolute differences between predictions and actual observations. Like RMSE, a smaller MAE suggests better performance.

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\mathcal{X}^i - \mathcal{Y}^i| \tag{10}$$

- **Grid Prediction Accuracy (GridACC).** GridACC is a domain-specific metric that measures the accuracy of predictions on a discretized grid (common in spatial data applications). It's the percentage of the number of correct grid cell predictions N_{acc} over the total number of grid cells N_{grid}.

$$\text{GridACC} = 100\% \times \frac{N_{\text{acc}}}{N_{\text{grid}}} \tag{11}$$

Each metric offers a unique perspective on imputation quality; RMSE and MAE focus on the magnitude of errors, Grid-ACC on the accuracy specific to spatial resolution. By employing all these metrics, we can obtain a comprehensive understanding of a model's performance across different dimensions of accuracy.

B. Result and Performance Analysis

Table II show the performance comparison between STIOS and the selected baseline methods on two real-world datasets. Specifically, we compare the performance of the best models after training based on the three evaluation metrics described in Section V-A4 . We first compare the models with deterministic models, leading to observations in Table II, where the best results are highlighted in bold font, and the second-best results are underlined.

We can observe that simply using traditional methods such as KNN or Transformer is not sufficient to handle the task of imputing spatio-temporal trajectory data under the condition of open environment. The BRTIS method underperforms, and these methods demonstrate unstable performance across different tasks. Moreover, sequence-based approaches (BRTIS, SAITS), aside from Transformer, demonstrate markedly better performance in the imputation tasks than simple traditional methods. Clearly, our model STIOS outperforms all non-generative baseline models, showing strong superiority in trajectory imputation.

Moreover, as can be seen from the Table II, our model STIOS achieves performance with strong competitiveness on

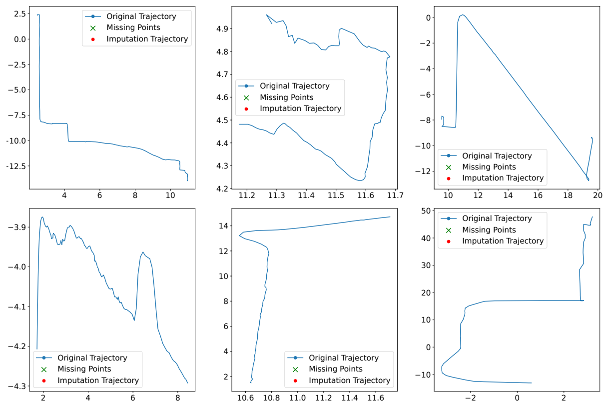

Fig. 4: Example of probabilistic time series imputation. Each subplot shows the probabilistic imputation of a trajectory. The horizontal and vertical axes represent the transformed coordinate values.

all metrics in three tasks across 2 real-world datasets, maintaining stable results across datasets of different sizes. It outperforms in the majority of metrics and is only second to CSDI in a few. This confirms the effectiveness of our model. We note that for the task at hand, most time series imputation models perform poorly in terms of trajectory grid prediction accuracy (GridACC). In fact, GridACC is an intuitive reflection of the completeness of trajectory data, representing the model's precision in a coarse-grained manner. However, the subpar performance of most models underscores the challenge of the problem, suggesting that most models treat trajectories as ordinary time series and disregard spatio-temporal information. Particularly, models that introduce structured approaches underperform in trajectory data. This may be due to structured state-space models not being able to learn the trajectory data effectively.

C. Probabilistic Imputation Visualization

As mentioned earlier, one of the advantages of applying probabilistic models to multivariate imputation is that they can generate various plausible imputations, which is beneficial for uncertainty estimation. We provide a visual example of imputation by STIOS on the Geolife dataset in "Fig. 4" on next page. Blue circles represent the original trajectory values, red circles represent the imputed values, and green crosses signify the missing values. To generate the visualization of probabilistic imputation, we produce plausible imputations 100 times. As shown in the figure, in most cases, our model fits the original time series well, and these results suggest that STIOS is capable of generating accurate estimations for missing values with high precision.

VI. Conclusion

In this work, we present STIOS for open environment scenarios. As a current state-of-the-art generative modelling approach, STIOS combines deterministic models with conditional diffusion models to effectively address spatio-temporal correlation and uncertainty. Tests on two real-world datasets show that STIOS outperforms existing state-of-the-art baseline models in a variety of datasets under different scenarios, and especially exhibits high competitiveness and superiority in open ocean and urban land scenarios. In addition, we propose a self-supervised training method inspired by masked language modelling that divides the data into conditional information and estimation targets. We believe that the proposed technique is a very promising generative approach in the field of spatio-temporal trajectory data. It offers the possibility of recovering and generating important trajectories and marks a step forward in the field. In our next work, we will continue to investigate trajectory imputation models for open scenarios in more situations and hope to propose more appropriate models for obtaining road grids information.

References

[1] M. Qi, J. Qin, Y. Wu, and Y. Yang, "Imitative non-autoregressive modeling for trajectory forecasting and imputation," in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun 2020.

[2] T. Xia, Y. Qi, J. Feng, F. Xu, F. Sun, D. Guo, and Y. Li, "Attnmove: History enhanced trajectory recovery via attentional network." *Proceedings of the AAAI Conference on Artificial Intelligence*, p. 4494–4502, Sep 2022.

[3] H. Sun, C. Yang, L. Deng, F. Zhou, F. Huang, and K. Zheng, "Periodicmove: Shift-aware human mobility recovery with graph neural network," in *Proceedings of the 30th ACM International Conference on Information & Knowledge Management*, Oct 2021.

[4] Y. Huang, H. Bi, Z. Li, T. Mao, and Z. Wang, "Stgat: Modeling spatial-temporal interactions for human trajectory prediction," in *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, Oct 2019.

[5] H. Jeon, J. Choi, and D. Kum, "Scale-net: Scalable vehicle trajectory prediction network under random number of interacting vehicles via edge-enhanced graph convolutional neural network," in *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Oct 2020.

[6] G. Zerveas, S. Jayaraman, D. Patel, A. Bhamidipaty, and C. Eickhoff, "A transformer-based framework for multivariate time series representation learning," in *Proceedings of the 27th ACM SIGKDD conference on knowledge discovery & data mining*, 2021, pp. 2114–2124.

[7] J. Jiang, D. Pan, H. Ren, X. Jiang, C. Li, and J. Wang, "Self-supervised trajectory representation learning with temporal regularities and travel semantics," in *2023 IEEE 39th international conference on data engineering (ICDE)*. IEEE, 2023, pp. 843–855.

[8] P. Felsen, P. Lucey, and S. Ganguly, *Where Will They Go? Predicting Fine-Grained Adversarial Multi-agent Motion Using Conditional Variational Autoencoders*, Jan 2018, p. 761–776.

[9] M. Qi, J. Qin, A. Li, Y. Wang, J. Luo, and L. Van Gool, *stagNet: An Attentive Semantic RNN for Group Activity Recognition*, Jan 2018, p. 104–120.

[10] S. Suwajanakorn, S. M. Seitz, and I. Kemelmacher-Shlizerman, "Synthesizing obama," *ACM Transactions on Graphics*, p. 1–13, Aug 2017.

[11] S. Taylor, T. Kim, Y. Yue, M. Mahler, J. Krahe, A. G. Rodriguez, J. Hodgins, and I. Matthews, "A deep learning approach for generalized speech animation," *ACM Transactions on Graphics*, p. 1–11, Aug 2017.

[12] M. Qi, W. Li, Z. Yang, Y. Wang, and J. Luo, "Attentive relational networks for mapping images to scene graphs," *Cornell University - arXiv,Cornell University - arXiv*, Nov 2018.

[13] M. Qi, Y. Wang, J. Qin, and A. Li, "Ke-gan: Knowledge embedded generative adversarial networks for semi-supervised scene parsing," in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun 2019.

[14] X. Wang, H. Zhang, P. Wang, Y. Zhang, B. Wang, Z. Zhou, and Y. Wang, "An observed value consistent diffusion model for imputing missing values in multivariate time series," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 2409–2418.

[15] J. Alcaraz and N. Strodthoff, "Diffusion-based time series imputation and forecasting with structured state space models," *arXiv preprint*, Aug 2022.

[16] M. Liang, B. Yang, R. Hu, Y. Chen, R. Liao, S. Feng, and R. Urtasun, *Learning Lane Graph Representations for Motion Forecasting*. Springer, Jan 2020, p. 541–556.

[17] Y. Liu, R. Yu, S. Zheng, E. Zhan, and Y. Yue, "Naomi: Non-autoregressive multiresolution sequence imputation," *Cornell University - arXiv,Cornell University - arXiv*, Jan 2019.

[18] B. Yang, G. Yan, P. Wang, C.-Y. Chan, X. Song, and Y. Chen, "A novel graph-based trajectory predictor with pseudo-oracle," *IEEE Transactions on Neural Networks and Learning Systems*, p. 7064–7078, Dec 2022.

[19] M. Berglund, T. Raiko, M. Honkala, L. Kärkkäinen, A. Vetek, and J. Karhunen, "Bidirectional recurrent neural networks as generative models," *Neural Information Processing Systems,Neural Information Processing Systems*, Dec 2015.

[20] Z. Che, S. Purushotham, K. Cho, D. Sontag, and Y. Liu, "Recurrent neural networks for multivariate time series with missing values," *Scientific Reports*, Apr 2018.

[21] W. Cao, D. Wang, J. Li, H. Zhou, L. Li, and Y. Li, "Brits: Bidirectional recurrent imputation for time series," *Neural Information Processing Systems,Neural Information Processing Systems*, May 2018.

[22] T.-Y. Choi, J. Kang, and J.-H. Kim, "Rdis: Random drop imputation with self-training for incomplete time series data," *Cornell University - arXiv,Cornell University - arXiv*, Oct 2020.

[23] Q. Suo, W. Zhong, G. Xun, J. Sun, C. Chen, and A. Zhang, "Glima: Global and local time series imputation with multi-directional attention learning," in *2020 IEEE International Conference on Big Data (Big Data)*, Dec 2020.

[24] W. Du, D. Côté, and Y. Liu, "Saits: Self-attention-based imputation for time series," *Expert Systems with Applications*, vol. 219, p. 119619, 2023.

[25] K. Rasul, C. Seward, I. Schuster, and R. Vollgraf, "Autoregressive denoising diffusion models for multivariate probabilistic time series forecasting," in *International Conference on Machine Learning*. PMLR, 2021, pp. 8857–8868.

[26] M. Salvaris, D. Dean, and W. H. Tok, *Generative Adversarial Networks*, Jan 2018, p. 187–208.

[27] P. Wang, C. Zhu, X. Wang, Z. Zhou, G. Wang, and Y. Wang, "Inferring intersection traffic patterns with sparse video surveillance information: An st-gan method," *IEEE Transactions on Vehicular Technology*, p. 9840–9852, Sep 2022.

[28] Y. Luo, Y. Zhang, X. Cai, and X. Yuan, "E²gan: End-to-end generative adversarial network for multivariate time series imputation," in *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence*, Aug 2019.

[29] X. Miao, Y. Wu, J. Wang, Y. Gao, X. Mao, and J. Yin, "Generative semi-supervised learning for multivariate time series imputation," *Proceedings of the AAAI Conference on Artificial Intelligence*, p. 8983–8991, Sep 2022.

[30] N. Chen, Y. Zhang, H. Zen, R. Weiss, M. Norouzi, and W. Chan, "Wavegrad: Estimating gradients for waveform generation," May 2021.

[31] Y. Song, J. Sohl-Dickstein, D. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-based generative modeling through stochastic differential equations," *International Conference on Learning Representations,International Conference on Learning Representations*, May 2021.

[32] Y. Tashiro, J. Song, Y. Song, and S. Ermon, "Csdi: Conditional score-based diffusion models for probabilistic time series imputation," *Cornell University - arXiv,Cornell University - arXiv*, Dec 2021.

[33] A. T. Hudak, N. L. Crookston, J. S. Evans, D. E. Hall, and M. J. Falkowski, "Nearest neighbor imputation of species-level, plot-scale forest structure attributes from lidar data," *Remote Sensing of Environment*, p. 2232–2245, May 2008.

[34] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017.

[35] V. Fortuin, D. Baranchuk, G. Rätsch, and S. Mandt, "Gp-vae: Deep probabilistic time series imputation," *Cornell University - arXiv,Cornell University - arXiv*, Jul 2019.

[36] J. Yoon, J. Jordon, and M. Schaar, "Gain: Missing data imputation using generative adversarial nets," *Cornell University - arXiv,Cornell University - arXiv*, Jun 2018.